Battery safety: swollen batteries and what to do

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Summary

This article describes how to identify a swollen or failing laptop battery, the immediate safety actions users should take, and the steps Desktop Support follows to quarantine, replace, and verify the battery. It covers the typical symptoms reported by users, common causes, and the self-help checks that can be performed before contacting the service desk.

Symptoms

- Battery icon shows 0% or does not charge
- Battery drains unusually fast (e.g., <30 minutes)
- Physical bulge or deformation of the battery pack visible on the underside of the laptop
- Laptop shuts down suddenly even when plugged in
- Charger LED does not illuminate or flickers
- Error message in Windows: "Battery not detected" or "Plugged in, not charging"

Common causes

- Aging lithium-ion cell beyond its design life (~2-3 years)
- Exposure to high temperatures (e.g., left in a hot vehicle or near vents)
- Manufacturing defect or internal short circuit
- Physical impact or puncture to the battery pack
- Using non-approved chargers that over-voltage the battery

Safety Precautions

If you suspect the battery is swollen, stop using the laptop immediately. Do not attempt to puncture, bend, or apply pressure to the battery. Disconnect the charger and shut down the system. Place the laptop on a non-flammable surface away from paper, fabrics, or other combustibles. Keep the area ventilated and avoid exposing the device to direct sunlight or heat sources until it can be inspected.

Self-Help Procedure - Check for Battery Swelling

Follow these steps to safely inspect your laptop for a swollen battery before contacting the Service Desk. Perform the check only when the laptop is powered off and unplugged.

1. Shut down the laptop completely (Start > Power > Shut down).
2. Disconnect the AC adapter and any peripheral cables.
3. Turn the laptop over so the bottom panel is facing up.
4. Visually examine the battery area (usually a rectangular block along the rear edge). Look for any bulging, warping, or separation between the battery and the chassis.
5. If the laptop has a removable battery (some older models), gently release the latch and lift the battery out - do not force it. Inspect the battery cells for swelling or leakage.
6. If you observe any deformation, bulging, fluid, or unusual odor, do not re-install the battery. Proceed to the next step.
7. Place the laptop in a safe location and contact the IT Service Desk via phone ext. 4357 (HELP) or the Teams channel "IT Help" for further instructions.

Service Desk Procedure - Desktop Support

Agents should treat any report of a swollen battery as a potential safety hazard. The following steps guide the agent from initial triage through resolution and verification.

1. Create a ticket and set the priority to P2 (High) due to safety risk; if the user reports smoke, fire, or imminent danger, escalate to P1 immediately.
2. Advise the user to power off the device, disconnect the charger, and keep the laptop in a safe, ventilated area (refer to the Safety Precautions section).
3. If the user is on-site, arrange for a Desktop Support technician to collect the laptop within the same business day. For

remote users, ship a pre-paid, padded return box with instructions to keep the device powered off and upright.
4. Upon receipt, inspect the battery in an ESD-safe area. Confirm swelling, leakage, or damage. If confirmed, quarantine the laptop: label it "Swollen Battery - Do Not Use" and store it in a designated fire-resistant cabinet.
5. Order a replacement battery from the approved spare parts inventory (Dell/Lenovo/HP/Apple as appropriate). Record the serial numbers of both the defective and replacement units for asset tracking.
6. Install the new battery, reconnect the charger, and perform a full charge-discharge cycle to validate health (>80% design capacity). Run the built-in battery diagnostics (e.g., Dell BIOS battery test, Lenovo Vantage, HP Support Assistant, or Apple System Report).
7. Update the asset record with the new battery details, close the ticket, and notify the user that the laptop is ready for pickup or delivery.
8. If the battery appears healthy but the user still experiences charging issues, proceed to charger troubleshooting (see KB0010038).

Verification and Aftercare

After the battery replacement, both the user and the agent should confirm that the device operates normally and that no safety concerns remain.

1. Ask the user to power on the laptop and verify that the battery charges to 100% and holds charge for at least 4 hours of typical use.
2. Confirm that the battery health percentage shown in the OS settings or vendor utility is above 80%.
3. Ensure the laptop does not exhibit abnormal heating during charging or use.
4. Remind the user to avoid exposing the laptop to extreme temperatures and to use only the supplied AC adapter or an approved replacement.
5. Close the ticket with resolution description: "Swollen battery identified, laptop quarantined, replacement battery installed and validated."

When to escalate

If the user reports smoke, sparks, fire, or a strong chemical odor from the battery, treat the incident as a P1 Critical issue. Immediately advise the user to evacuate the area, disconnect power if safe to do so, and call the IT Service Desk via phone ext. 4357 (HELP). The Service Desk will then escalate to the Security Operations team and the Facilities Safety team for emergency response. For all other swollen-battery cases, follow the Desktop Support procedure above; escalation to the Hardware Engineering team is only required if repeated failures occur on the same model line, in which case include the asset tag, battery serial number, and photos of the swelling in the escalation note.

Keywords: battery swollen, laptop not charging, battery drains fast, charger not working, battery health, replace battery, swollen battery safety, quarantine laptop, Desktop Support, battery replacement, power issue, battery bulge